

COMP3242 Week 1 Practice Exam (Solutions)

Q1

- (a) Flattened shape: $(16, 3 \cdot 32 \cdot 32) = (16, 3072)$.
- (b) For mapping $3072 \rightarrow 100$: weight shape $(100, 3072)$, bias shape $(100,)$.
- (c) Output shape: $(16, 100)$.

Q2

$$Ax = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 2 + 0 \\ -2 - 3 + 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad y = Ax + b = \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

Q3

- (a) $u^T v = 1 \cdot 3 + (-2) \cdot 0 + 2 \cdot (-1) = 1$.
- (b) $\|u\|_2 = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{9} = 3$.

Q4

PyTorch aligns trailing dimensions. $b \in \mathbb{R}^4$ is treated as shape $(1, 4)$ and broadcast across the first dimension of $X \in \mathbb{R}^{8 \times 4}$. Therefore $X + b$ is valid and output shape is $(8, 4)$.

Q5

Bug: ‘CrossEntropyLoss’ expects class indices of integer type (‘torch.long’), not floating labels.
Correct fix:

```
labels = labels.long()
loss = torch.nn.CrossEntropyLoss()(logits, labels)
```